



COMPUTER AWARENESS

- On receiving an interrupt from an I/O device, the CPUs
 - hand over the control of address and data bus to interrupting device.
 - branch off to interrupt service subroutine immediately.
 - branch off to interrupt service subroutine after completion of current instruction.
 - None of the above
- Micro-programmed control unit is
 - faster than hard-wired unit.
 - slower than hard-wired unit.
 - to facilitate easy implementation of new instructions.
 - both (B) and (C).
- Index register in a digital computer is used for
 - pointing to the stack address.
 - indirect addressing.
 - keeping track the number of times loop is executed.
 - address modification.
- In the virtual memory system, the address space specified by address lines of the CPU must be....than the physical memory size and.....than the secondary storage size.
 - smaller, smaller
 - smaller, larger
 - larger, smaller
 - larger, larger
- The switching expression corresponding to $f(A, B, C, D) = \sum(1, 4, 5, 9, 11, 12)$ is
 - $B\bar{C}\bar{D} + \bar{A}\bar{C}D + A\bar{B}D$
 - $AB\bar{C} + ACD + \bar{B}\bar{C}D$
 - $AC\bar{D} + \bar{A}B\bar{C} - A\bar{C}\bar{D}$
 - $\bar{A}BD + ACD + B\bar{C}\bar{D}$
- Assuming all numbers are in 2's complement representation, which of the following numbers is divisible by 11111011?
 - 11100100
 - 11010111
 - 11011011
 - 00000110
- A switching circuit that produces one in a set of input bits as an output based on the control value of control bits is termed as
 - Full Adder
 - Inverter
 - Multiplexer
 - Converter
- A Computer with a 32 bit word size uses 2's complement to represent numbers. The range of integers that can be represented by this computer is
 - -2^{32} to 2^{32}
 - -2^{31} to 2^{32}
 - -2^{31} to $2^{31} - 1$
 - -2^{32} to 2^{31}
- To change upper case to the lower case letter in ASCII, correct mask and operation should be
 - 0100000 and NOR.
 - 0100000 and NAND.
 - 0100000 and OR.
 - None of the above
- Why is the width of a data bus so important to the processing speed of a computer?
 - The narrower it is, the greater the computer's processing speed.
 - The wider it is, the more data can fit into the main memory.
 - The wider it is, the greater the computer's processing speed.
 - The wider it is, the slower the computer's processing speed.

ANSWER

1.	C	2.	A	3.	B	4.	C	5.	A
6.	A	7.	C	8.	C	9.	C	10.	b

SOLUTION

- (c)
Branch off to interrupt service subroutine after completion of current instruction.
- (a)
Faster than hard-wired unit.
- (b) Indirect addressing.
- (c) Larger, smaller.
- (a)
For the blue group
C and D maintain the same state but A and B change.
C is zero (0) and has to be negated before it can be included. Thus, the first term is
 $ABCD + \bar{A}\bar{B}\bar{C}\bar{D} = \bar{A}\bar{C}D(B + \bar{B}) = \bar{A}\bar{C}D$
For the green group
Second term is $\bar{A}\bar{B}\bar{C}D + AB\bar{C}\bar{D} = (A + \bar{A})\bar{B}\bar{C}D = \bar{B}\bar{C}D$
- (a) 1's complement of 11111011 is
Now its 2's complement = 00000100
$$\begin{array}{r} 00000100 \\ + \quad \quad 1 \\ \hline 00000101 \end{array}$$

Its decimal equivalent
 $= 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 4 + 0 + 1 = 5$
From option 0.
1's complement of 00000110 is = 11111001
Now, its 2's complement = 11111001
$$\begin{array}{r} 11111001 \\ + \quad \quad 1 \\ \hline 11111010 \end{array}$$

Its decimal equivalent
 $= 1 \times 2^7 + 1 \times 2^6 + 1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$
 $= 128 + 64 + 32 + 16 + 8 + 2 = 250$
Which is divisible by 5.
- (c) Multiplexer.
- (c) Range of 2's complement with n bits is
 $-(2^{n-1})$ to $2^{n-1} - 1$
- (c) Mask is 0100000 with operation OR.
- (b) Wider the data bus more will be memory capacity.

	CD	00	01	11	10
AB	00		1		
	01	1	1		
	11	1			
	10		1	1	